

SQL databases, which are ACID properties compliant.

NewSQL is a contemporary relational database management system, which provides the same scalable performance as NoSQL systems for online transaction processing (OLTP) read-write workloads, and maintains the ACID guarantees of a classical database system.

Nowadays, Web applications use data in heterogeneous formats, which includes audio, video, text, streaming content, signals, images, pixels and many others. In each file, there are a number of file formats. For example, in video, there are a number of file formats including MPEG, MP4, AVI, 3GP, WMV, OGG, FLV and others. In the same manner, image or graphics file formats include GIF, PNG, JPEG, PCX, BMP, TIFF and many others.

Now the major issue is the compatibility of the Web application with all these file formats in different domains. It is at this point that implementing NoSQL databases makes the most sense. In NoSQL databases, any type of file format can be processed and integrated in the Web application.

NoSQL databases provide a storage and retrieval system that is different from the tabular relations used in relational databases. The data structure in NoSQL databases is entirely different from the classical RDBMS. NoSQL databases are rapidly being used in Big Data and realtime Web applications.

There have been various approaches to classifying NoSQL databases, each with different categories and sub-categories. Because of the variety of approaches and the way they overlap, it is difficult to get a clear overview of non-relational databases.

NoSQL Databases for Multiple Domains

Column: Accumulo, Cassandra, Druid, HBase

Document: Clusterpoint, Apache CouchDB, Couchbase, MarkLogic, MongoDB

Key-Value: Dynamo, FoundationDB, MemcacheDB, Redis, Riak, FairCom-treeACE, Aerospike

Graph: Allegro, Neo4J, InfiniteGraph, OrientDB, Virtuoso, Stardog

MongoDB is one of the prominent cross-platform document-oriented NoSQL databases released under a combination of the GNU Affero GPL and the Apache Licence. It is free and open source software.

According to media reports, the database technology used for the back-end of the Aadhaar cards in India is MongoDB. Aadhaar implementation involves enrolling thousands of Indians each day and adds terabytes of data to the data repository. MongoDB as well as other data management and analytics software providers continue to produce insights that aid Aadhaar and the lives of millions of India's citizens.

MongoDB's customers include companies such as Disney, the New York Times, Cisco, MTV, Forbes, Craigslist, ADP, AstraZeneca, FourSquare, IBM, Intuit, Microsoft, McAfee, Under Armour, and eBay.

Initially developed by 10gen (now MongoDB Inc) in

October 2007 as a component of a planned platform (as a service product), the organisation shifted to an open source development model in 2009, with 10gen offering commercial support and other services. Since then, MongoDB has been adopted by a number of major websites and services.

Web application development using PHP and NoSQL databases

In order to use MongoDB with PHP, we need the MongoDB PHP driver.

Download the driver from the URL [Download PHP Driver](#).

Now unzip the archive and use `php_mongo.dll` in your PHP extension directory ("ext" by default) and add the following line to your `php.ini` file:

```
extension=php_mongo.dll
```

Database connection and selecting the database

To make a connection, we need to specify the database's name. If it doesn't exist, then MongoDB creates it automatically.

```
<?php
// connect to mongodb
$m = new MongoClient();
echo "Connection to database successfully";
// select a database
$db = $m->mydb;
echo "Database mydb selected";
?>
```

When the program is executed, it will give the following result:

```
Connection to database successfully
Database mydb selected
```

Creating a collection

To create a collection, type:

```
<?php
// connect to mongodb
$m = new MongoClient();
echo "Connection to database successfully";
// select a database
$db = $m->mydb;
echo "Database mydb selected";
$collection = $db->createCollection("mycol");
echo "Collection created successfully";
?>
```